DISTRICT POPULATION ANALYSIS AND THEMATIC MAPPING PROJECT

A brief project report presenting the analysis and geographical visualization of population data for five selected districts in Uganda.

1. INTRODUCTION

This project demonstrates how Python programming and Geographic Information Systems (GIS) can be used to analyse and present district population data. It focuses on Wakiso, Kampala, Mukono, Jinja and Masaka, combining population statistics with geographical boundary information to produce a thematic map, a comparison chart and a statistical report.

2. PROBLEM STATEMENT

Population figures presented only as tables can make geographical differences between districts difficult to interpret. The project addresses this by combining population data with district boundaries and presenting the results through thematic mapping and graphical comparison.

3. PROJECT OBJECTIVES

General objective: To develop a Python-based system for analysing selected district population data and presenting the results geographically.

Specific objectives: (i) load district population data from a CSV file; (ii) load Uganda district boundaries from a GeoJSON file; (iii) process and match the population and boundary data; (iv) calculate summary statistics; and (v) generate a thematic map and a population comparison chart.

4. METHODOLOGY

The system was developed in Python using PyCharm. Population figures for the five selected districts were organized in a CSV file, while geographical boundaries were supplied in GeoJSON format. The program loads the two datasets, checks and matches district names, processes the population values, calculates summary statistics, and uses the resulting data to produce a district population thematic map and a bar chart.

5. RESULTS

The analysis produced a total population of 6,150,000 and an average of 1,230,000 for the five selected districts. Wakiso recorded the highest population (2,500,000), while Masaka recorded the lowest (350,000). The following pages present the statistical summary, thematic map and comparison chart.

6. CHALLENGES ENCOUNTERED AND SOLUTIONS

Data-file location: Files were placed in the appropriate project folders and file paths were checked.
Boundary loading: The boundary dataset format and file path were checked, and the GeoJSON file was loaded successfully.
District-name matching: Differences between names in the CSV and boundary dataset were investigated and adjusted.
Map presentation: The map was refined to include a legend, north arrow and scale bar.

GROUP 17 MEMBERS

NAME	REGISTRATION NUMBER
KAWENJA ROBERT	25/1/324/D/1707
KIBIRANGO ANTHONY IAN	25/1/324/D/999
KAWEESI SHAFIK KHALID	25/1/324/D/052
WALUUBE PATRICIA LYTZ R.	25/1/324/D/952
OKUMU TRAVIS KAZAA	25/1/324/D/1924

DISTRICT POPULATION ANALYSIS REPORT

SUMMARY INDICATOR	RESULT
Total Population	6,150,000
Average Population	1,230,000
Highest Population District	Wakiso
Lowest Population District	Masaka

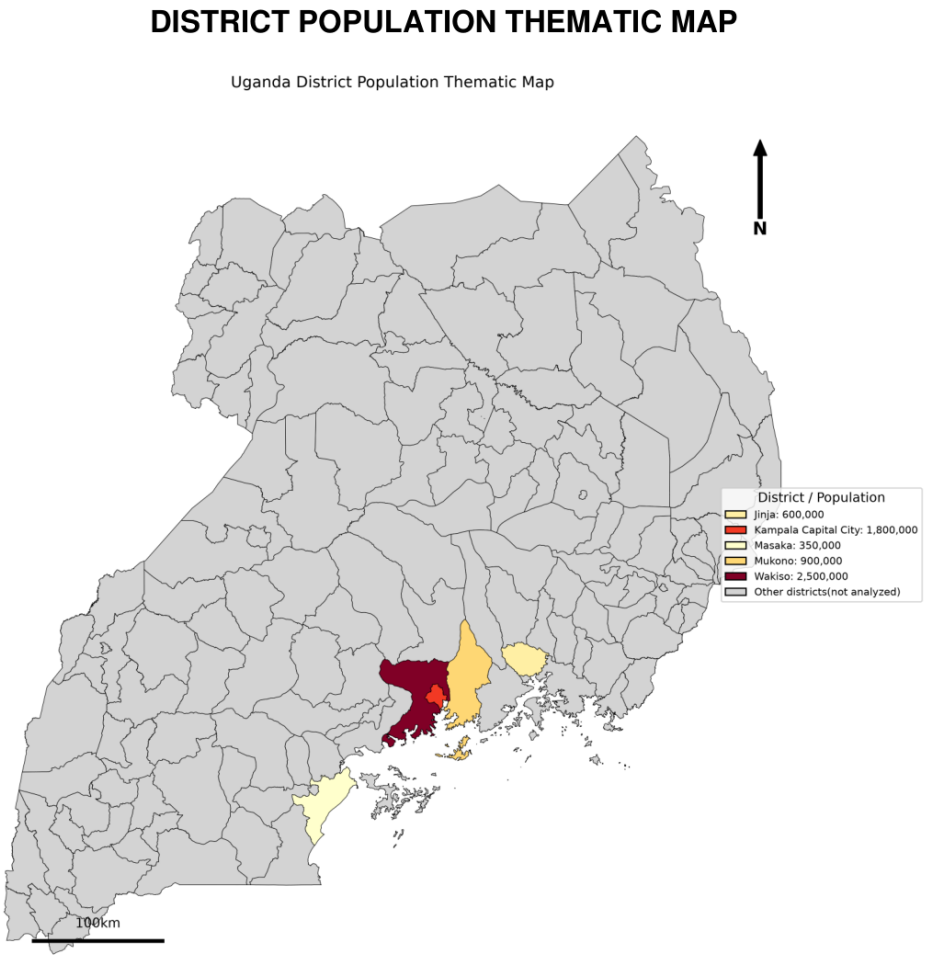
Population by Selected District

District	Population
Wakiso	2,500,000
Kampala	1,800,000
Mukono	900,000
Jinja	600,000
Masaka	350,000

The figures above summarize the five districts included in this project.

DISTRICT POPULATION THEMATIC MAP

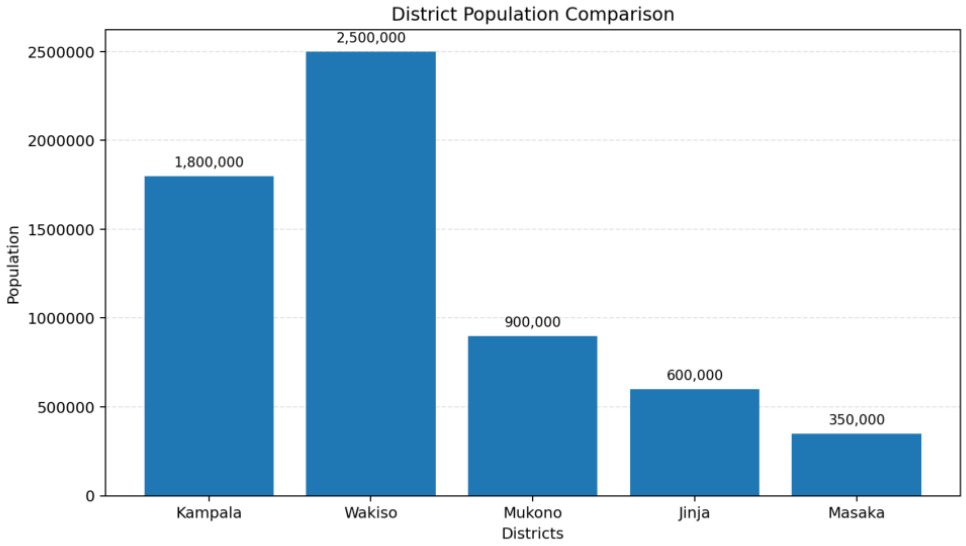
Geographical distribution of the five selected district population figures



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DISTRICT POPULATION COMPARISON

Population values for the five selected districts



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PROJECT CONCLUSIONS AND RECOMMENDATIONS

PROJECT CONCLUSIONS

1. The total population of the five selected districts is 6,150,000.
2. The average population across the selected districts is 1,230,000.
3. Wakiso has the highest population among the selected districts.
4. Masaka has the lowest population among the selected districts.

RECOMMENDATIONS

1. Use population data to guide public service planning.
2. Consider population differences when allocating resources.
3. Regularly update geographical and population data.
4. Use thematic maps to support informed planning decisions.